NACA 9412 Calculations

- **>Defining the "NACA 9412" Geometry**
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- Using the NACA 4-digit convention, we define our shape parameters:
- **1st Digit (Max Camber, m): 9%** (0.09). This provides good lift.
- **2nd Digit (Max Camber Location, p): 40%** (0.4). The peak curvature is 0.4m from the front.
- **3rd/4th Digits (Thickness, t): 12%** (0.12). This is relatively thick, good for structural strength and fuel storage.
- **Chord Length (c): 1.0 meter** (for easy calculation).
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Step 1: Calculate the Thickness (yt)

- *The "Flesh"*
- We need a shape that is rounded at the front (to catch air from different angles) and razor-sharp at the back (to reduce drag). To get this specific "teardrop" shape, we use a standard polynomial equation.
- Basically, for any point x, this formula tells you how much "meat" to add to the bone:
- $yt = 5t\ [\ 0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4\ ]$

Step 2: Calculate the Camber (yc)

- *The "Spine"*
- Now we need to bend the wing. We don't use a single curve for this; we actually splice two different parabolas together.
- **The Front (0 to 40%):** We use one equation to curve the nose up to the peak.
- **The Back (40% to 100%):** We use a different equation to gently slope it back down to the tail.
- Since our peak (p) is at **0.4**, you just check where you are (x) and pick the right formula:
- **If you are in the Front (x<0.4):**
- $yc = (m / p^2)\ (\ 2px - x^2\ )$
- **If you are in the Back (x≥0.4):**
- $yc = (m / (1 - p)^2)\ [\ (1 - 2p) + 2px - x^2\ ]$

Step 3: Find the Slope (dyc /dx) & Angle (θ)

- *The "Perpendicular Push"*
- Here is the tricky part where most people get stuck.
- You can't just add the thickness straight up and down. Since the spine is curved, "up" changes direction as you move along the wing. We need to add the thickness **perpendicular** to the spine.
- To do that, we first need the slope of our spine. Just like the camber, the slope formula changes depending on if you are in the front or back:
- **Front Section (0 ≤ x < p)**
- $dyc/dx = (2m / p^2)\ (\ p - x\ )$
- **Rear Section (p ≤ x ≤ 1)**
- $dyc/dx = (2m / (1 - p)^2)\ (\ p - x\ )$

- Once we have the slope, we turn it into an angle (θ) using arctan:
- $\theta = \arctan(dyc/dx)$
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Step 4: The Final Coordinates

- *Putting it all together*
- Now we have everything we need. We take our point on the spine (x,yc) and push out by the thickness amount (yt) at our calculated angle (θ).
- **For the Upper Surface (Top of the wing):** We subtract a bit of x and add to y (tilting it back and up):
- $xU = x - yt \sin(\theta)$
- $yU = yc + yt \cos(\theta)$
- **For the Lower Surface (Bottom of the wing):** We add a bit of x and subtract from y (tilting it forward and down):
- $xL = x + yt \sin(\theta)$
- $yL = yc - yt \cos(\theta)$

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		m(max camber)=	0.09							
		p(max position)=	0.4							
		t(thickness)=	0.12							
		AOA(α)=	-10							
		Conversion(Radians)=	-0.174532925199433							
X	Yc	Chord X	Chord Y	Slope angle(θ)	angle(θ)	Yt	xu	yu	xl	yl
0	0	0	0	0.45	0.42285392613	0	0	0	0	0
0.0025	0.00112148437	1	0	0.4471875	0.42051258882	0.00871668416293359	-0.00105839836608194	0.00907876933059903	0.00605839836608194	-0.00683580058059903
0.005	0.0022359375			0.444375	0.41816633839	0.0122131474844947	0.000040418234808682	0.0133967403471253	0.00995958176519132	-0.00892486534712534
0.0075	0.00334335937			0.4415625	0.41581518527	0.0148485818137621	0.00150212643866142	0.0169266540645141	0.0134978735613386	-0.0102399353145141
0.01	0.00444375			0.43875	0.41345914017	0.02	0.00315485487507699	0.0200452203701949	0.016845145124923	-0.0111577203701949
0.0125	0.00553710937			0.4359375	0.41109821402	0.0189390266528367	0.00493165657758868	0.0228981838923593	0.0200683434224113	-0.0118239651423593
0.015	0.0066234375			0.433125	0.40873241805	0.0206367117634073	0.00679801150418467	0.0255602074759084	0.0232019884958153	-0.0123133324759084
0.0175	0.00770273437			0.4303125	0.40636176374	0.0221790089180859	0.00873330325652294	0.02807559536202	0.0262666967434771	-0.01267012661202
0.02	0.008775			0.4275	0.40398626283	0.02	0.0107240268904997	0.0304731827122813	0.0292759731095003	-0.0129231827122813
0.0225	0.00984023437			0.4246875	0.40160592735	0.0249151289048086	0.0127607509421196	0.0327729768218118	0.0322392490578804	-0.0130925080718118
0.025	0.0108984375			0.421875	0.39922076957	0.0261471986425573	0.0148365706727315	0.0349895292387106	0.0351634293272685	-0.0131926542387106
0.0275	0.01194960937			0.4190625	0.39683080205	0.0273061512090666	0.0169462444558895	0.0371338135071056	0.0380537555441105	-0.0132345947571056
0.03	0.01299375			0.41625	0.39443603760	0.03	0.0190856769232694	0.0392143459801335	0.0409143230767306	-0.0132268459801335
0.0325	0.01403085937			0.4134375	0.39203648932	0.0294406064581788	0.0212515923716478	0.0412378921873408	0.0437484076283522	-0.0131761734373408
0.035	0.0150609375			0.410625	0.38963217058	0.030429738573433	0.0234413191651974	0.0432099319226547	0.0465586808348026	-0.0130880569226547
0.0375	0.01608398437			0.4078125	0.38722309501	0.0313738751622281	0.0256526421888019	0.0451349767089723	0.0493473578111981	-0.0129670079589723
0.04	0.0171			0.405	0.38480927654	0.03	0.0278836987749979	0.0470167931481534	0.0521163012250021	-0.0128167931481534
0.0425	0.01810898437			0.4021875	0.38239072935	0.0331433512990601	0.0301329033826079	0.0488585641540636	0.0548670966173921	-0.0126405954040636
0.045	0.0191109375			0.399375	0.37996746791	0.033975300974609	0.0323988918619619	0.0506630079551815	0.0576011081380381	-0.0124411329551815
0.0475	0.02010585937			0.3965625	0.37753950697	0.0347757057471129	0.034680479409215	0.0524324676417548	0.060319520590785	-0.0122207488917548
0.05	0.02109375			0.39375	0.37510686155	0.04	0.0369766283044736	0.0541689797029241	0.0630233716955264	-0.0119814797029241
0.0525	0.02207460937			0.3909375	0.37266954695	0.0362907640219678	0.0392864227818568	0.0558743272791234	0.0657135772181432	-0.0117251085291234
0.055	0.0230484375			0.388125	0.37022757877	0.0370092021471848	0.0416090491903064	0.0575500820982442	0.0683909508096936	-0.0114532070982442
0.0575	0.02401523437			0.3853125	0.36778097286	0.03770374801983	0.0439437801412855	0.0591976379012662	0.0710562198587145	-0.0111671691512662
0.06	0.024975			0.3825	0.36532974536	0.04	0.046289961703505	0.0608182373764574	0.073710038296495	-0.0108682373764574
0.0625	0.02592773437			0.3796875	0.36287391275	0.0390266537475586	0.048647002956291	0.0624129940785956	0.076352997043709	-0.0105575253285956
0.065	0.0268734375			0.376875	0.36041349171	0.0396574192370084	0.0510143673900321	0.0639829104286047	0.078985632609679	-0.0102360354286047
0.0675	0.02781210937			0.3740625	0.35794849923	0.0402691402266691	0.053391565768506	0.0655288926170893	0.081608434231494	-0.00990467386708929
0.07	0.02874375			0.37125	0.35547895263	0.04	0.0557781501595134	0.0670517630383479	0.0842218498404866	-0.00956426303834786
0.0725	0.02966835937			0.3684375	0.35300486945	0.0414391261885181	0.0581737089075847	0.0685522707369413	0.0868262910924153	-0.00921555198694132
0.075	0.0305859375			0.365625	0.35052626757	0.0419990347204226	0.0605778623726337	0.0700311002415146	0.0894221376273663	-0.00885922524151457
0.0775	0.03149648437			0.3628125	0.34804316514	0.0425432040345356	0.0629902592961364	0.0714888790798782	0.0920097407038636	-0.00849591032987 82
0.08	0.0324			0.36	0.34555558058	0.04	0.0654105736850971	0.0729261842080637	0.0945894263149029	-0.00812618420806371
0.0825	0.03329648437			0.3571875	0.34306353262	0.0435869351331478	0.0678385021260867	0.0743435475390617	0.0971614978739133	-0.00775057878906169
0.085	0.0341859375			0.354375	0.34056704027	0.0440876809518246	0.0702737614587121	0.0757414607205656	0.0997262385412879	-0.00736958572056556
0.0875	0.03506835937			0.3515625	0.33806612283	0.0445750655553144	0.0727160867512182	0.0771203792826462	0.102283913248782	-0.00698366053264616
0.09	0.03594375			0.34875	0.33556079990	0.05	0.0751652295314408	0.0784807262539332	0.104834770468559	-0.00659322625393325
0.0925	0.03681210937			0.3459375	0.33305109136	0.0455116869781123	0.0776209562346722	0.0798228953271671	0.107379043765328	-0.00619867657716711
0.095	0.0376734375			0.343125	0.33053701737	0.0459618122088822	0.0800830468366738	0.0811472536408414	0.109916953163326	-0.00580037864084144
0.0975	0.03852773437			0.3403125	0.32801859841	0.0464003588846681	0.0825512936454391	0.0824541442322955	0.112448706354561	-0.00539867548229553
0.1	0.039375			0.3375	0.32549585523	0.0468277042382395	0.0850255002296555	0.0837438882084281	0.114974499770344	-0.00499388820842807
0.11	0.04269375			0.32625	0.31536206558	0.05	0.0949783391997613	0.0887371547516896	0.125021660800239	-0.00334965475168964
0.12	0.0459			0.315	0.30516082461	0.05	0.105013179251584	0.0934772087251306	0.134986820748416	-0.00167720872513064
0.13	0.04899375			0.30375	0.29489360074	0.05	0.115121239521601	0.0979773235914359	0.144878760478399	0.000010176408564102
0.14	0.051975			0.2925	0.28456193706	0.05	0.125295065883123	0.102248279032057	0.154704934116877	0.00170172096794258
0.15	0.05484375			0.28125	0.27416745111	0.05	0.13552821512398	0.106298985114739	0.16447178487602	0.00338851488526095
0.16	0.0576			0.27	0.26371183446	0.05	0.145815031364813	0.110136920871061	0.174184968635187	0.00506307912893866
0.17	0.06024375			0.25875	0.25319685204	0.06	0.156150483742368	0.113768450512588	0.183849516257632	0.00671904948741182
0.18	0.062775			0.2475	0.24262434140	0.06	0.166530046005656	0.117199056542803	0.193469953994344	0.00835094345719707
0.19	0.06519375			0.23625	0.23199621166	0.06	0.176949605528727	0.120433514957769	0.203050394471273	0.00995398504223092
0.2	0.0675			0.225	0.22131444234	0.0573754299023625	0.187405393436067	0.123476029173037	0.212594606563933	0.0115239708269634
0.21	0.06969375			0.21375	0.21058108192	0.06	0.197893930182032	0.126330333943709	0.222106069817968	0.0130571660562908
0.22	0.071775			0.2025	0.19979824624	0.06	0.208411982641542	0.128999777078806	0.231588017358458	0.0145502229211936
0.23	0.07374375			0.19125	0.18896811669	0.06	0.21895652990969	0.131487384459137	0.24104347009031	0.0160001155408633
0.24	0.0756			0.18	0.17809293823	0.06	0.229524735782611	0.133795912318826	0.250475264217389	0.0174040876811738
0.25	0.07734375			0.16875	0.16717501709	0.06	0.240113926428501	0.135927889682954	0.259886073571499	0.0187596103170457
0.26	0.078975			0.1575	0.15621671844	0.06	0.250721572136003	0.137885653104742	0.269278427863997	0.0200643468952579
0.27	0.08049375			0.14625	0.14522046374	0.06	0.261345272298149	0.139671375311801	0.278654727701851	0.0213161246881986
0.28	0.0819			0.135	0.13418872795	0.06	0.271982742987058	0.141287088984755	0.288017257012942	0.0225129110152447
0.29	0.08319375			0.12375	0.12312403655	0.06	0.282631806619704	0.142734706608451	0.297368193380296	0.023652793391549
0.3	0.084375			0.1125	0.11202896242	0.0600172663939703	0.293290383323332	0.144016037125934	0.306709616676668	0.0247339628740665
0.31	0.08544375			0.10125	0.10090612246	0.06	0.303956483690767	0.145132799967728	0.316043516309233	0.0257547000322716
0.32	0.0864			0.09	0.08975817416	0.06	0.314628202678146	0.146086636909489	0.325371797321854	0.0267133630905107
0.33	0.08724375			0.07875	0.07858781204	0.06	0.325303714445582	0.146879122119594	0.334696285554418	0.0276083778804059
0.34	0.087975			0.0675	0.06739776371	0.06	0.335981267978583	0.147511770687654	0.344018732021417	0.0284382293123459
0.35	0.08859375			0.05625	0.05619078620	0.06	0.346659183357318	0.147986045869908	0.353340816642682	0.0292014541300917
0.36	0.0891			0.045	0.04496661815	0.06	0.357335848564004	0.148303365244353	0.362664151435996	0.0298966347556468
0.37	0.08949375			0.03375	0.03373719429	0.06	0.368009716737215	0.148465105934377	0.371990283262785	0.0305223940656234
0.38	0.089775			0.0225	0.02249620427	0.06	0.378679303796767	0.148472609032571	0.381320696203233	0.0310773909674285
0.39	0.08994375			0.01125	0.01124952542	0.06	0.389343186374984	0.14832718333472	0.390656813625016	0.0315603166652803
0.4	0.09			0	0	0.058030108476479	0.4	0.148030108476479	0.4	0.031969891523521
0.42	0.0899			-0.01	-0.00999966666	0.06	0.420572226179181	0.147122617918066	0.419427773820819	0.0326773820819336
0.44	0.0896			-0.02	-0.01999733339	0.06	0.441125913805488	0.145895690274416	0.438874086194512	0.033304309725584
0.46	0.0891			-0.03	-0.02999100048	0.06	0.461657717029967	0.144357234332223	0.458342282970033	0.0338427656677773
0.48	0.0884			-0.04	-0.03997868711	0.05	0.482164580482132	0.14251451205329	0.477835419517868	0.0342854879467104
0.5	0.0875			-0.05	-0.04995839577	0.0529402520005716	0.502643710025318	0.140374200506353	0.497356289974682	0.0346257994936466
0.52	0.0864			-0.06	-0.05992815515	0.05	0.523092546537454	0.137942442290903	0.516907453462546	0.0348575577090966
0.54	0.0851			-0.07	-0.06988600165	0.05	0.543508742086267	0.13522488694667	0.536491257913733	0.0349751130533303
0.56	0.0836			-0.08	-0.07982998577	0.05	0.56389013801972	0.132226725246499	0.55610986198028	0.0349732747535012
0.58	0.0819			-0.09	-0.08975817417	0.05	0.584234744604947	0.128952717832748	0.575765255395053	0.0348472821672524
0.6	0.08			-0.1	-0.09966865247	0.0456336906586779	0.604540721933192	0.125407219331923	0.595459278066808	0.0345927806680773
0.62	0.0779			-0.11	-0.10955952675	0.04	0.624806361872123	0.121594198837482	0.615193638127877	0.0342058011625178
0.64	0.0756			-0.12	-0.11942892605	0.04	0.645030070895741	0.117517257464506	0.634969929104259	0.0336827425354944
0.66	0.0731			-0.13	-0.12927500400	0.04	0.66521035365979	0.113179643536843	0.65478964634021	0.0330203564631567
0.68	0.0704			-0.14	-0.13909594147	0.04	0.685345797219888	0.108584265856346	0.674654202780112	0.0322157341436542
0.7	0.0675			-0.15	-0.14888994767	0.0366390671267802	0.705435055812528	0.103733705416855	0.694564944187472	0.0312662945831453
0.72	0.0644			-0.16	-0.15865526215	0.03	0.725476836137122	0.0986302258570154	0.714523163862878	0.0301697741429846
0.74	0.0611			-0.17	-0.16839015715	0.03	0.745469883091529	0.0932757828913459	0.734530116908471	0.0289242171086541
0.76	0.0576			-0.18	-0.17809293825	0.03	0.765412965924735	0.0876720329151939	0.754587034075265	0.0275279670848061
0.78	0.0539			-0.19	-0.18776194655	0.03	0.785304864779326	0.0818203409438211	0.774695135220674	0.0259796590561789
0.8	0.05			-0.2	-0.19739555985	0.026231179804725	0.805144357603417	0.0757217880170873	0.794855642396583	0.0242782119829127
0.82	0.0459			-0.21	-0.20699219425	0.02	0.824930207417309	0.0693771781776618	0.815069792582691	0.0224228218223382
0.84	0.0416			-0.22	-0.21655030495	0.02	0.844661149924489	0.0627870451113131	0.835338850075511	0.0204129548886869
0.86	0.0371			-0.23	-0.22606838795	0.02	0.864335881460015	0.0559516585218043	0.855664118539985	0.0182483414781957
0.88	0.0324			-0.24	-0.23554498075	0.02	0.883953047271905	0.048871030299603	0.876046952728095	0.0159289697003 97
0.9	0.0275			-0.25	-0.24497866315	0.0144771727147185	0.903511230133123	0.0415449205324928	0.896488769866877	0.0134550794675072
0.92	0.0224			-0.26	-0.25436805855	0.01	0.923008939283187	0.0339728433968736	0.916991060716813	0.0108271566031264
0.94	0.0171			-0.27	-0.26371183445	0.01	0.942444599699392	0.0261540729607122	0.937555400300608	0.00804592703928776
0.96	0.0116			-0.28	-0.27300870305	0.01	0.96181654169831	0.0180876489225352	0.95818345830169	0.00511235107746477
0.98	0.00590000000			-0.29	-0.28225742195	0.00	0.981122990868539	0.00977238230530826	0.978877009131461	0.00202761769469178
1	0			-0.3	-0.29145679445	0.00126	1.00036205833581	0.00120686111937865	0.999637941664186	-0.00120686111937865